BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EEE 304 (July 2024)

Digital Electronics Laboratory

Final Project Report

Section: B1 Group: 07

Digital Voting Machine

Course Instructors:

Nafis Sadik
Lecturer
Dept. of EEE, BUET

H M Tahmeed Ul Islam
Lecturer (PT)
Dept. of EEE, BUET

Signature of Instructor: _____

Academic Honesty Statement:

"In signing this statement, we hereby certify that the work on this project is our own and that we have not copied the work of any other students (past or present), and cited all relevant sources while completing this project. We understand that if we fail to honor this agreement, we will each receive a score of ZERO for this project and be subject to failure of this course."

Signature: _____ **Full Name:** Rafatul Islam Tamim **Student ID:** 2006083	**Signature:** _____ **Full Name:** Iffat Islam Pinky **Student ID:** 2006084
Signature: _____ **Full Name:** Tabassum Shikder Promy **Student ID:** 2006086	**Signature:** _____ **Full Name:** Nahian Tasnim **Student ID:** 2006098

Table of Contents

1 Abstract 1

2 Introduction 1

3 Design 2

3.1 Problem Formulation (PO(b)) 2

3.1.1 Identification of Scope 2

3.1.2 Literature Review 2

3.1.3 Formulation of Problem 3

3.1.4 Analysis 3

3.2 Design Method (PO(a)) 4

3.3 Circuit Diagram 7

3.4 Hardware Design 9

3.5 Full Source Code of Firmware 11

4 Implementation 14

4.1 Description 14

5 Design Analysis and Evaluation 15

5.1 Novelty 15

5.2 Design Considerations (PO(c)) 15

5.2.1 Considerations to public health and safety 15

5.2.2 Considerations to environment 15

5.2.3 Considerations to cultural and societal needs 16

5.3 Investigations (PO(d)) 17

5.4 Limitations of Tools (PO(e)) 17

5.5 Impact Assessment (PO(f)) 17

5.5.1 Assessment of Societal and Cultural Issues 17

5.5.2 Assessment of Health and Safety Issues 18

5.5.3 Assessment of Legal Issues 19

5.6 Ethical Issues (PO(h)) 19

6 Reflection on Individual and Team work (PO(i)) 20

6.1 Individual Contribution of Each Member 20

6.2 Mode of TeamWork 20

6.3 Diversity Statement of Team 20

6.4 Log Book of Project Implementation 21

7 Communication to External Stakeholders (PO(j)) 21
7.1 Executive Summary 21
7.2 User Manual 22
7.3 Drive Link of Project Code 22
7.4 YouTube Link 23
8 Project Management and Cost Analysis (PO(k)) 23
8.1 Bill of Materials 23
8.2 Timeline of Project Implementation 24
9 Future Work (PO(l)) 24
10 References 25

1 Abstract

The digitalization of voting systems is critical for improving the efficiency, security, and transparency of electoral processes. This project focuses on the design and implementation of a Digital Voting Machine (DVM) using Field-Programmable Gate Arrays (FPGAs), a technology known for its high performance, reliability, and flexibility. The FPGA-based DVM is developed to address challenges such as vote tampering, manual counting errors, and system inefficiencies prevalent in traditional voting systems. The design includes modules for voter authentication, secure vote casting, real-time tallying, and result verification. By leveraging the parallel processing capabilities and reconfigurability of FPGAs, this system ensures rapid computation, robust security, and scalability for future enhancements. This report highlights the system architecture, hardware-software integration, and performance analysis, demonstrating the viability of FPGA-based solutions in modernizing electoral systems.

2 Introduction

Elections are a fundamental pillar of democracy, and the integrity of the voting process is essential to maintain public trust. Traditional voting methods, including paper ballots and electronic voting machines (EVMs), have been subject to criticisms related to security vulnerabilities, inefficiency, and lack of transparency. The integration of advanced digital technologies, such as Field-Programmable Gate Arrays (FPGAs), offers a promising solution to address these issues.

FPGAs provide a versatile platform for designing high-speed, secure, and customizable systems. In this project, an FPGA-based Digital Voting Machine (DVM) is proposed to modernize the voting process. The FPGA's reconfigurable architecture allows for real-time processing of votes, secure encryption mechanisms, and tamper-proof designs, making it an ideal candidate for implementing a secure voting system.

This report presents the development of an FPGA-based DVM, covering key aspects such as system design, functional modules, and security protocols. The machine incorporates features like voter authentication through unique identification, real-time vote counting, and result storage with encryption. By ensuring high-speed processing, fault tolerance, and ease of deployment, this FPGA-based system aims to overcome the limitations of existing voting technologies. The introduction of such a system could significantly enhance the transparency, reliability, and scalability of electoral processes, paving the way for widespread adoption in democratic institutions.

3 Design

3.1 Problem Formulation (PO(b))

3.1.1 Identification of Scope

The scope of the Digital Voting Machine (DVM) Project implemented on FPGA encompasses the design, development, and verification of a secure, efficient, and reliable system for electronic voting. This project aims to utilize the reconfigurable and parallel-processing capabilities of Field Programmable Gate Arrays (FPGAs) to achieve accurate and real-time voting functionalities, ensuring robustness and ease of use. The DVM will incorporate a user-friendly interface through input mechanisms such as push buttons or switches, enabling voters to cast their votes for specific candidates. To provide immediate feedback, the system will utilize output devices such as LEDs, 7-segment displays, or LCD screens to confirm successful voter registration and to display real-time results.

The scope assumes that the DVM will operate in an environment where voting occurs in an organized manner, with each voter allowed to cast only one vote. However, the system will have limitations, such as support for a limited number of candidates and the absence of persistent storage, which means votes will not be retained after a power reset. To improve usability, a manual reset mechanism will allow for a new voting session after results are displayed.

This project has practical applications in environments like small-scale elections, student councils, or organizational voting processes, where quick and reliable vote counting is essential. Additionally, the scope for future enhancements is significant. Biometric authentication systems such as fingerprint verification could be integrated to improve voter security. Non-volatile memory can be added to ensure votes are stored even during power interruptions, and graphical user interfaces could replace basic displays for better usability.

3.1.2 Literature Review

Electronic voting systems have become a focus of research due to the inefficiencies of traditional paper-based voting, which is prone to delays, human error, and security vulnerabilities. FPGA-based digital voting systems offer a promising alternative, providing speed, reliability, and enhanced security through hardware-based solutions. Patel et al. (2019) demonstrated that FPGA designs excel in real-time processing and accuracy, making them ideal for rapid and error-free vote counting. Similarly, Khan et al. (2020) emphasized the tamper-resistant nature of FPGA systems, which are less susceptible to hacking compared to software-based solutions.

A key concern in digital voting systems is ensuring voter authentication and preventing duplicate voting. Sharma et al. (2018) proposed integrating biometric authentication with FPGA logic, significantly enhancing security by ensuring only authorized voters can participate. However, this approach adds complexity to the hardware design. Input validation and button debouncing are also critical issues in such systems. Ahmad et al. (2017) addressed this by designing FPGA circuits with debounce filtering logic to ensure accurate voter registration, even in noisy real-world environments. Furthermore, Alam and Reddy (2016) demonstrated the flexibility of FPGA-based systems, which can be easily

customized for varying voting requirements, such as different numbers of candidates or additional features like vote encryption.

3.1.3 Formulation of Problem

Traditional voting systems suffer from various limitations:

- *Paper-based systems*: Prone to human errors and delays in counting.
- *Microcontroller-based systems*: Limited processing capability.
- Lack of security mechanisms to prevent multiple votes by a single voter.

This project addresses these issues by developing an FPGA-based digital voting machine that ensures:

- Secure voting through voter authentication.
- Efficient vote logging and counting.
- Real-time display of results.

3.1.4 Analysis

The need for a secure, reliable, and efficient voting system has become increasingly evident in modern electoral processes. Traditional paper-based voting systems are prone to errors, and delays in vote counting, often resulting in mistrust in election outcomes. Similarly, software-based electronic voting systems, while faster, are vulnerable to cybersecurity threats, tampering, and data corruption. These challenges highlight the necessity for a hardware-based solution that ensures the integrity, speed, and reliability of the voting process.

A key issue in existing voting systems is the lack of real-time vote tallying combined with adequate security measures. In many cases, vote counting is either manual or dependent on centralized systems prone to malfunctions or attacks. Furthermore, there is often insufficient protection against invalid voting, such as multiple votes by the same person. This calls for a system that can address these vulnerabilities while maintaining simplicity and cost-effectiveness for small-scale and controlled environments.

The use of Field Programmable Gate Arrays (FPGAs) for building a digital voting machine offers significant advantages. FPGAs are reconfigurable, allowing for the development of custom hardware solutions tailored to specific voting requirements. They ensure accurate and immediate vote counting. Additionally, hardware-based systems are inherently more secure than purely software-driven systems, as their logic cannot be easily altered during operation.

Another issue is the reliability of input handling in voting systems. Mechanical switches and buttons used for vote input can generate erroneous signals due to bounce effects, leading to inaccurate voter registration. Therefore, the inclusion of hardware debouncing mechanisms in the FPGA design is critical for ensuring error-free operation. Moreover, existing systems often lack a robust reset mechanism to initialize the voting process and ensure fairness before the start of each session.

3.2 Design Method (PO(a))

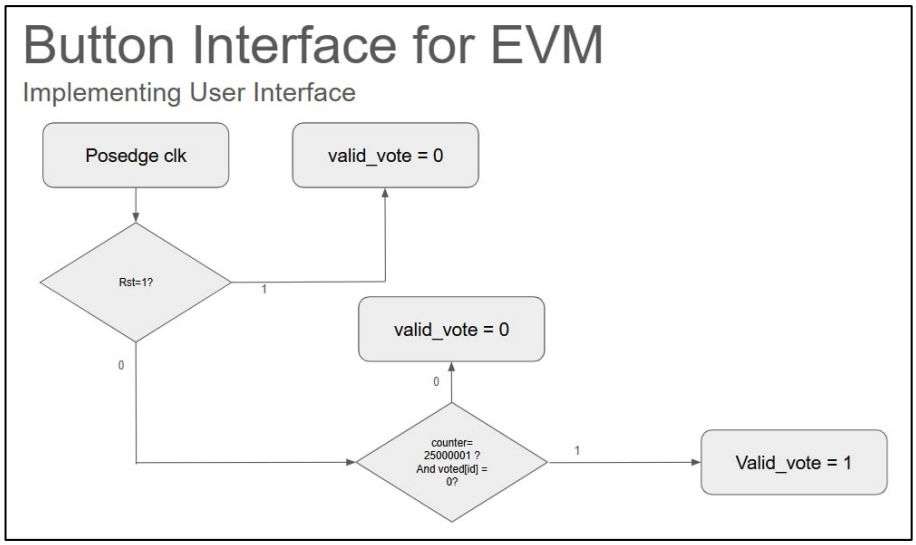
The system architecture comprises the following modules:

Top-Level Module (VotingMachine1)

- Integrates all submodules: `buttonControl`, `authentication`, `voteLogger`, `modeControl`, and `digitsDisplay`.
- Connects inputs such as `clock`, `reset`, `mode`, `button signals`, and `voter_id` to corresponding modules.
- Outputs the final vote counts to a 7-segment display.
- This module applies hierarchical design principles, ensuring seamless communication between submodules. Logical high and low signal states represent key control operations.

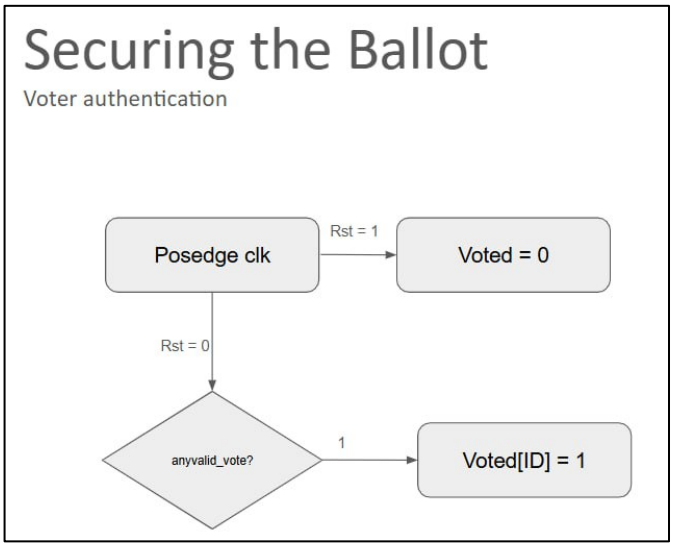
Button Control Module

- Debounces button inputs and validates long presses (25 million clock cycles).
- Prevents accidental votes caused by signal glitches.
- Implements timing logic using a counter that tracks 25 million clock cycles, corresponding to a 1-second button press for validation. This ensures input reliability and precision.
- Debounces button inputs and validates long presses (25 million clock cycles).
- Prevents accidental votes caused by signal glitches.



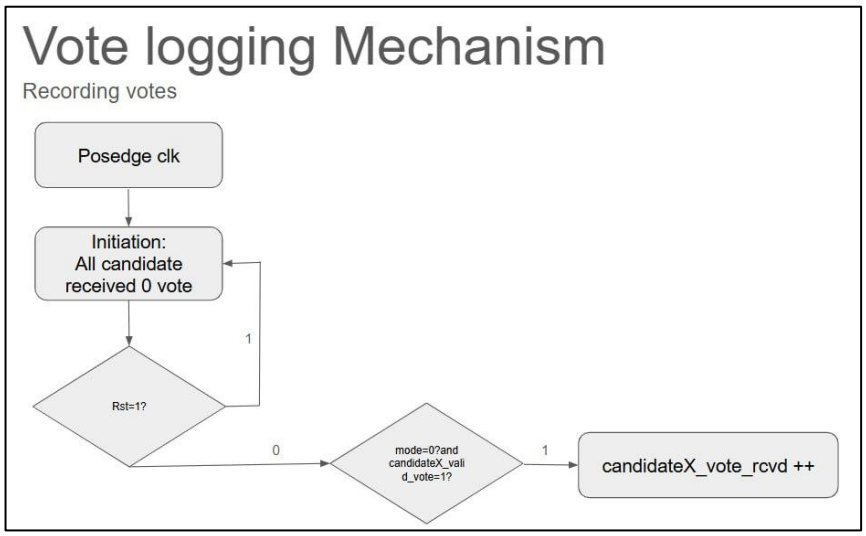
Authentication Module

- Maintains a `voted` register (256 bits) to ensure each voter ID can vote only once.
- Updates the register when a valid vote is cast.



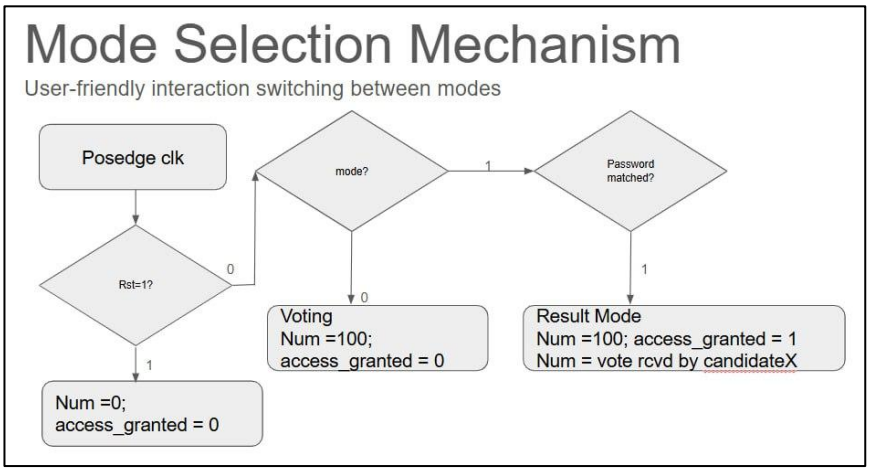
Vote Logger Module

- Logs votes for three candidates using counters.
- Increments counters upon receiving a valid vote signal.



Mode Control Module

- Manages system modes: `voting mode` and `result mode`.
- Displays the vote count for selected candidates during result mode.



Digits Display Module

- Converts numeric vote counts (0-99) into 7-segment display patterns.
- To generate the display output, the module performs division and modulus operations to separate the tens and ones place digits of the vote count. For example, given a vote count `num`, the tens place is calculated as `num / 10` and the ones place as `num % 10`. These computations ensure the correct digits are mapped to the corresponding 7-segment patterns for display.
- Displays tens and ones place of the vote count.

3.3 Circuit Diagram

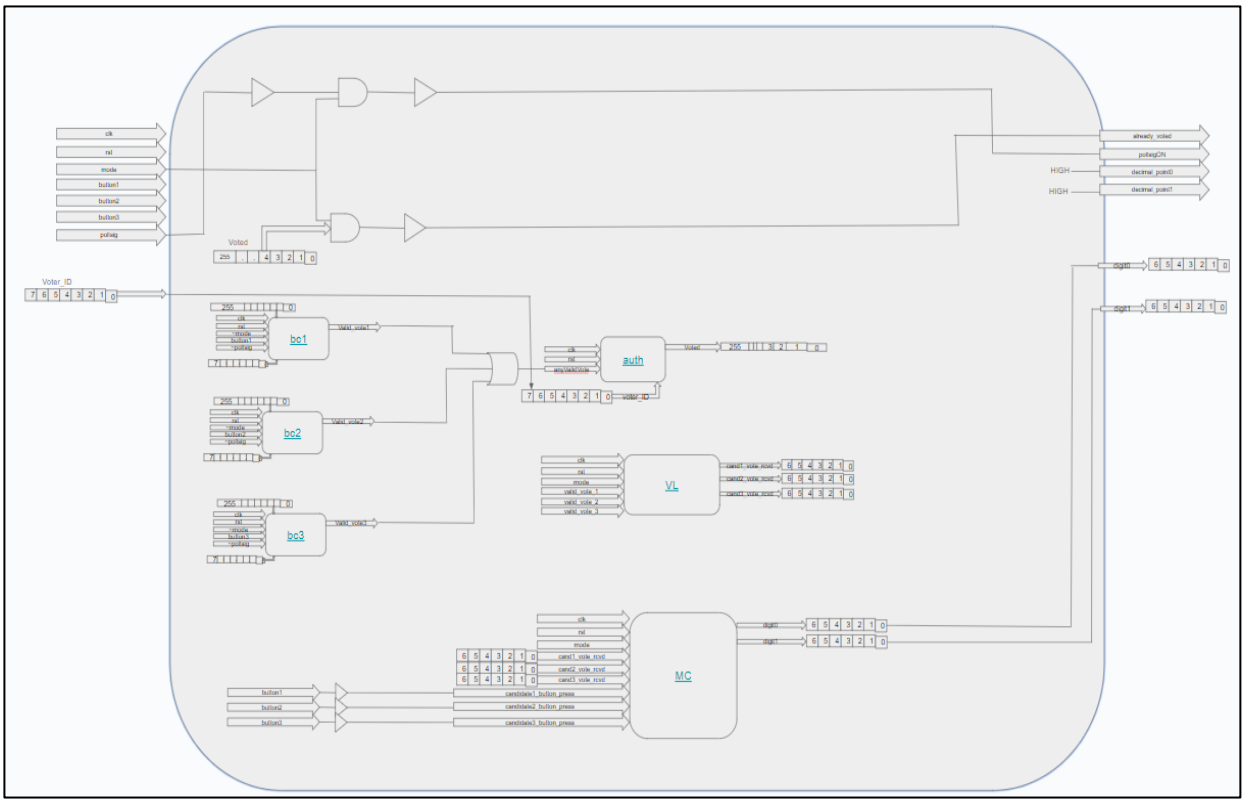


Fig: Integrated Diagram of the whole system

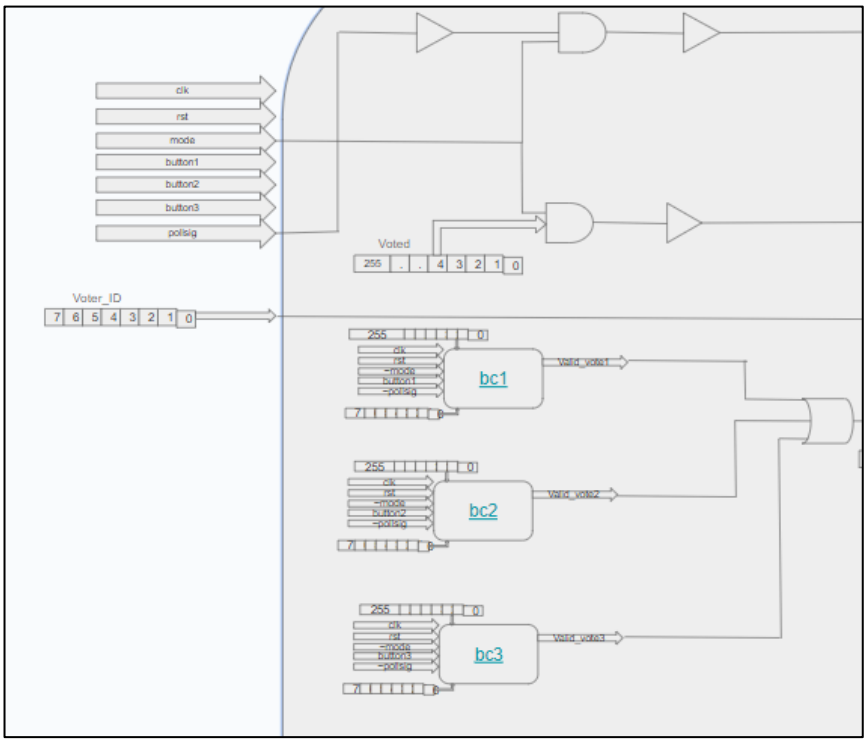


Fig: Focusing on the input and button control part

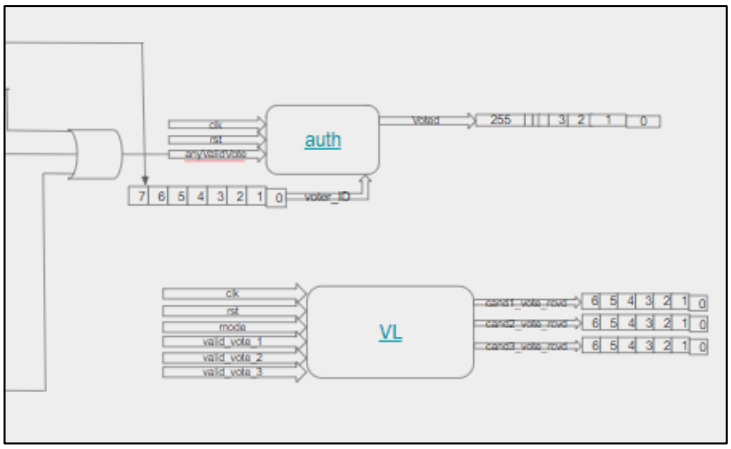


Fig: Focusing on the Authentication and Vote Logging part

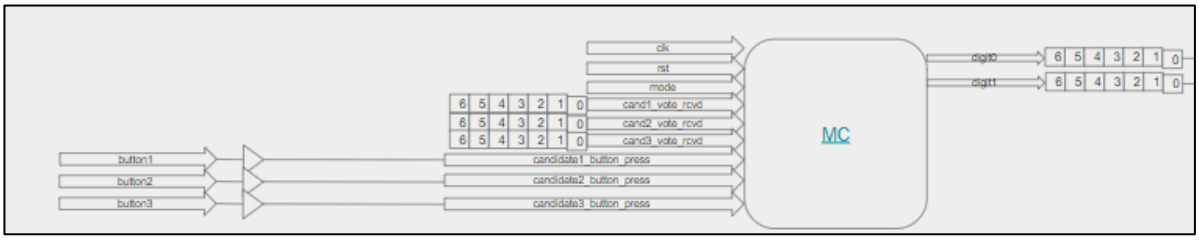


Fig: Focusing on the processing (result publishing after mode control)

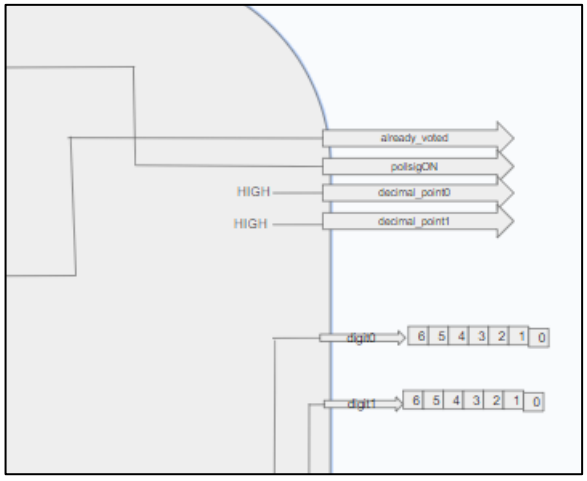
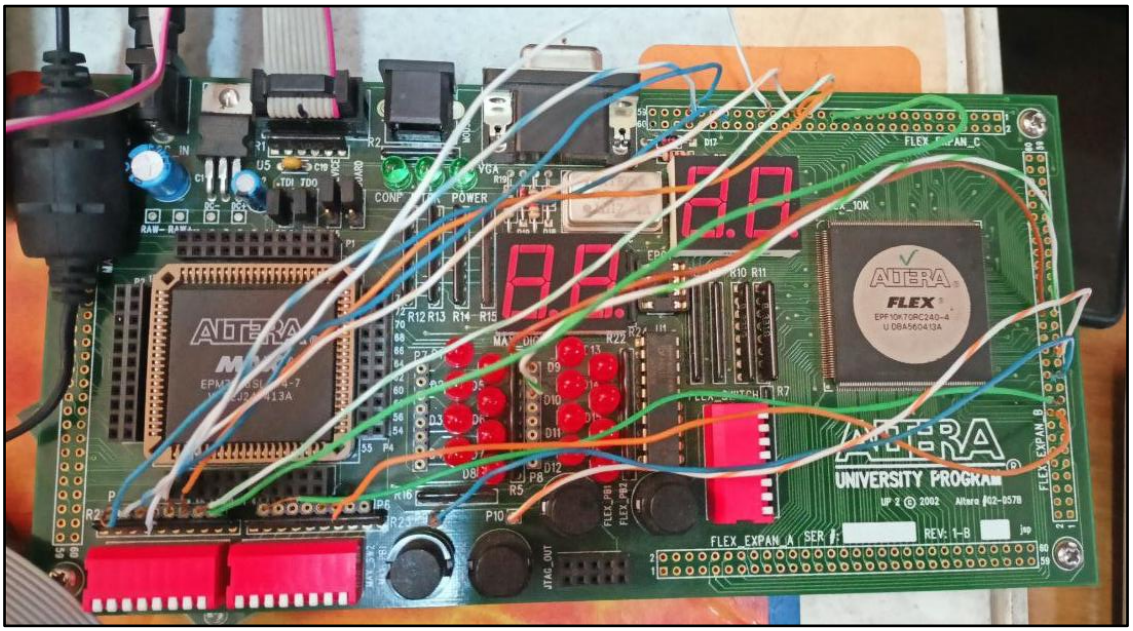


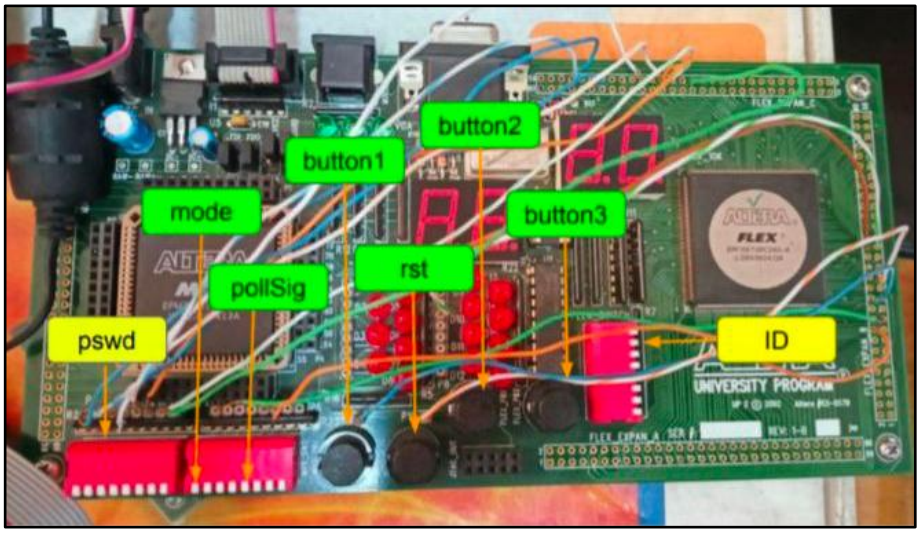
Fig: Focusing on the output result

3.4 Hardware Design

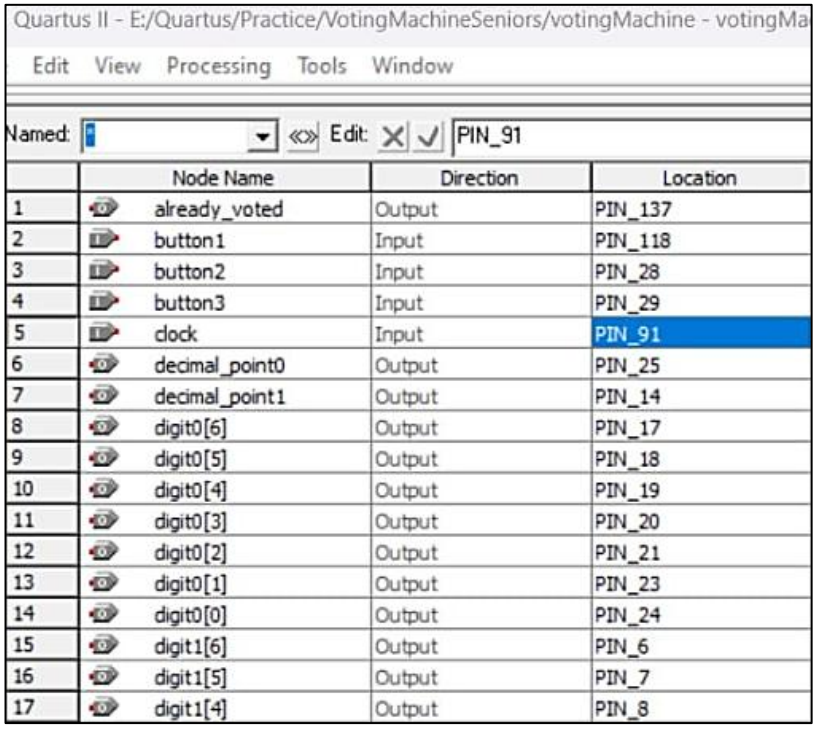
This is the picture of the Altera UP2 Development Board where we designed the digital voting machine.



In the Altera UP2 development board, MAX (EPM7128SLC84) and FLEX (EPF10K70RC240) are integrated. In the Quartus pin planner, we used only FLEX. So, for using the buttons under the part MAX of the board, jumpers were used to connect those buttons with the holes(pins) of FLEX. The pin planning was done following the datasheet of the board.



The picture of pin planner in Quartus is attached below:



	Node Name	Direction	Location
1	already_voted	Output	PIN_137
2	button1	Input	PIN_118
3	button2	Input	PIN_28
4	button3	Input	PIN_29
5	clock	Input	PIN_91
6	decimal_point0	Output	PIN_25
7	decimal_point1	Output	PIN_14
8	digit0[6]	Output	PIN_17
9	digit0[5]	Output	PIN_18
10	digit0[4]	Output	PIN_19
11	digit0[3]	Output	PIN_20
12	digit0[2]	Output	PIN_21
13	digit0[1]	Output	PIN_23
14	digit0[0]	Output	PIN_24
15	digit1[6]	Output	PIN_6
16	digit1[5]	Output	PIN_7
17	digit1[4]	Output	PIN_8
18	digit1[3]	Output	PIN_9
19	digit1[2]	Output	PIN_11
20	digit1[1]	Output	PIN_12
21	digit1[0]	Output	PIN_13
22	mode	Input	PIN_114
23	password[7]	Input	PIN_228
24	password[6]	Input	PIN_229
25	password[5]	Input	PIN_227
26	password[4]	Input	PIN_214
27	password[3]	Input	PIN_204
28	password[2]	Input	PIN_202
29	password[1]	Input	PIN_200
30	password[0]	Input	PIN_195
31	pollsig	Input	PIN_113
32	pollsigON	Output	PIN_136
33	reset	Input	PIN_127
34	voter_id[7]	Input	PIN_41
35	voter_id[6]	Input	PIN_40
36	voter_id[5]	Input	PIN_39
37	voter_id[4]	Input	PIN_38
38	voter_id[3]	Input	PIN_36
39	voter_id[2]	Input	PIN_35
40	voter_id[1]	Input	PIN_34
41	voter_id[0]	Input	PIN_33
42	<<new node>>		

3.5 Full Source Code of Firmware

```
`timescale 1ns / 1ps

module votingMachine(
  input clock,
  input reset,
  input mode,
  input button1,
  input button2,
  input button3,
  input pollsig,
  input [7:0] voter_id,
  input [7:0] password, // New input for password
  output already_voted,
  output pollsigON,
  output decimal_point0,
  output decimal_point1,
  output [6:0] digit0,
  output [6:0] digit1
);

  assign decimal_point0 = 1;
  assign decimal_point1 = 1;

  wire valid_vote_1;
  wire valid_vote_2;
  wire valid_vote_3;
  wire [6:0] cand1_vote_recvd;
  wire [6:0] cand2_vote_recvd;
  wire [6:0] cand3_vote_recvd;
  wire [255:0] voted;
  wire anyValidVote;

  wire reset_n;
  wire mode_n;
  wire button1_n;
  wire button2_n;
  wire button3_n;
  wire pollsig_n;
  wire [7:0] voter_id_n;

  assign reset_n = ~reset;
  assign mode_n = ~mode;
  assign button1_n = ~button1;
  assign button2_n = ~button2;
  assign button3_n = ~button3;
  assign pollsig_n = ~pollsig;
  assign voter_id_n = ~voter_id;

  assign pollsigON = ~(pollsig_n & mode);
  assign already_voted = ~(voted[voter_id_n] & mode);
  assign anyValidVote = valid_vote_1 | valid_vote_2 |
valid_vote_3 ;

  buttonControl bc1(
    clock(clock),
    reset(reset_n),
    button(button1_n),
    mode(mode_n),
    pollsig(pollsig_n),
    voter_id(voter_id_n),
    voted(voted),
    valid_vote(valid_vote_1)
  );

buttonControl bc2(
    clock(clock),
    reset(reset_n),
    button(button2_n),
    mode(mode_n),
    pollsig(pollsig_n),
    voter_id(voter_id_n),
    voted(voted),
    valid_vote(valid_vote_2)
  );

  buttonControl bc3(
    clock(clock),
    reset(reset_n),
    button(button3_n),
    mode(mode_n),
    pollsig(pollsig_n),
    voter_id(voter_id_n),
    voted(voted),
    valid_vote(valid_vote_3)
  );

authentication auth(
    clock(clock),
    reset(reset_n),
    anyValidVote(anyValidVote),
    voter_id(voter_id_n),
    voted(voted)
  );

  voteLogger VL(
    clock(clock),
    reset(reset_n),
    mode(mode_n),
    cand1_vote_valid(valid_vote_1),
    cand2_vote_valid(valid_vote_2),
    cand3_vote_valid(valid_vote_3),
    cand1_vote_recvd(cand1_vote_recvd),
    cand2_vote_recvd(cand2_vote_recvd),
    cand3_vote_recvd(cand3_vote_recvd)
  );

  modeControl MC(
    clock(clock),
    reset(reset_n),
    mode(mode_n),
    password(password), // New input for password
    valid_vote_casted(anyValidVote),
    candidate1_vote(cand1_vote_recvd),
    candidate2_vote(cand2_vote_recvd),
    candidate3_vote(cand3_vote_recvd),
    candidate1_button_press(button1_n),
    candidate2_button_press(button2_n),
    candidate3_button_press(button3_n),
    digit0(digit0),
    digit1(digit1)
  );

endmodule
```

Table: Source Code for the main module (Voting Machine)

```
`timescale 1ns / 1ps

module modeControl(
input clock,
input reset,
input mode,
input valid_vote_casted,
input [7:0] password, // New input for password
input [6:0] candidate1_vote,
input [6:0] candidate2_vote,
input [6:0] candidate3_vote,
input candidate1_button_press,
input candidate2_button_press,
input candidate3_button_press,
output [6:0] digit0,
output [6:0] digit1
);

reg [6:0] num;
reg access_granted;

initial begin num = 100;
access_granted = 0;
end

// Define the correct password (e.g.,8'b10101010 for
170 in decimal)
parameter [7:0] CORRECT_PASSWORD = 8'b10101010;

//mode0 -> voting mode, mode 1 -> result mode
```

```
always @(posedge clock)
begin
    if(reset)
    begin
        num = 0;
        access_granted = 0;
    end

    else if(~mode)
    begin              // mode = 0 -> voting mode
      num = 100;          // No digits displayed
      access_granted = 0;
    end

  else if(password == CORRECT_PASSWORD)
// mode = 1 -> result mode
    begin
      access_granted = 1;
// Grant access if password matches
      if(candidate1_button_press)
        num = candidate1_vote;
      else if(candidate2_button_press)
        num = candidate2_vote;
      else if(candidate3_button_press)
        num = candidate3_vote;
    end
  else
    begin
    access_granted = 0;
        num = 100;
// Display nothing if access denied
        end
end
//digitsDisplay
display_votes(.num(num), .digit0(digit0),
digit1(digit1));
digitsDisplay display_votes(num, digit0, digit1);

endmodule
```

Table: Source Code for Mode Control

```
`timescale 1ns / 1ps

module authentication(
  input clock,
  input reset,
  input anyValidVote,
  input [7:0] voter_id,
  output reg [255:0] voted
);

// might have the valid_votes and voted as inout

  always @(posedge clock)
  begin

  if (reset) voted = 0;
  else if (anyValidVote) voted[voter_id] = 1;

  end

endmodule
```

```
`timescale 1ns / 1ps

module voteLogger(
input clock,
input reset,
input mode,
input cand1_vote_valid,
input cand2_vote_valid,
input cand3_vote_valid,
output reg [6:0] cand1_vote_recvd,
output reg [6:0] cand2_vote_recvd,
output reg [6:0] cand3_vote_recvd
);

always @(posedge clock)
begin
    if(reset)
    begin
        cand1_vote_recvd <= 0;
        cand2_vote_recvd <= 0;
        cand3_vote_recvd <= 0;
    end
    else if(~mode)
    begin
        if(cand1_vote_valid)
            cand1_vote_recvd = cand1_vote_recvd + 1;
        else if(cand2_vote_valid)
            cand2_vote_recvd = cand2_vote_recvd + 1;
        else if(cand3_vote_valid)
            cand3_vote_recvd = cand3_vote_recvd + 1;
    end
end
endmodule
```

Table: Source Code for Authentication Module *Source code for Vote Logger*

```
`timescale 1ns / 1ps

module buttonControl(
input clock,
input reset,
input button,
input mode,
input pollsig,
input [7:0] voter_id,
input [255:0] voted,
output reg valid_vote
);
reg [24:0] counter;

always @(posedge clock)
begin
  if(reset | mode | ~pollsig)
        counter = 0;
    else
    begin
        if(button & counter < 25000001)
            counter = counter + 1;
        else if(!button)
            counter = 0;
    end
end

always @(posedge clock)
begin
    if(reset)
        valid_vote = 1'b0;
    else
    begin
        if(counter == 25000000 & ~voted[voter_id])
            valid_vote = 1'b1;
        else
            valid_vote = 1'b0;
    end
end
endmodule
```

```
`timescale 1ns / 1ps

module digitsDisplay (
  input [6:0] num,
    output reg [6:0] digit0, // 7-bit output for
segments a to g of digit 0
    output reg [6:0] digit1 // 7-bit output for
segments a to g of digit 1
);
wire [6:0] segment_data[0:9]; // Data for segments a to
g for digits 0 to 9

// Define 7-segment patterns for digits 0 to 9
assign segment_data[0] = 7'b0000001; // 0
assign segment_data[1] = 7'b1001111; // 1
assign segment_data[2] = 7'b0010010; // 2
assign segment_data[3] = 7'b0000110; // 3
assign segment_data[4] = 7'b1001100; // 4
assign segment_data[5] = 7'b0100100; // 5
assign segment_data[6] = 7'b0100000; // 6
assign segment_data[7] = 7'b0001111; // 7
assign segment_data[8] = 7'b0000000; // 8
assign segment_data[9] = 7'b0000100; // 9

// Display the number
always @(*) begin
    if (num >= 0 && num <= 99) begin
        digit1 = segment_data[num / 10]; // Set
segments for digit 1
        digit0 = segment_data[num % 10]; // Set
segments for digit 0
    end
  else begin
        digit0 = 7'b1111111; // Display 'Err' if input
is out of range
        digit1 = 7'b1111111;
    end
end
endmodule
```

Table: Source Code for Button Control *Source Code for Digit Display*

4 Implementation

4.1 Description

This is the flowchart of system design.

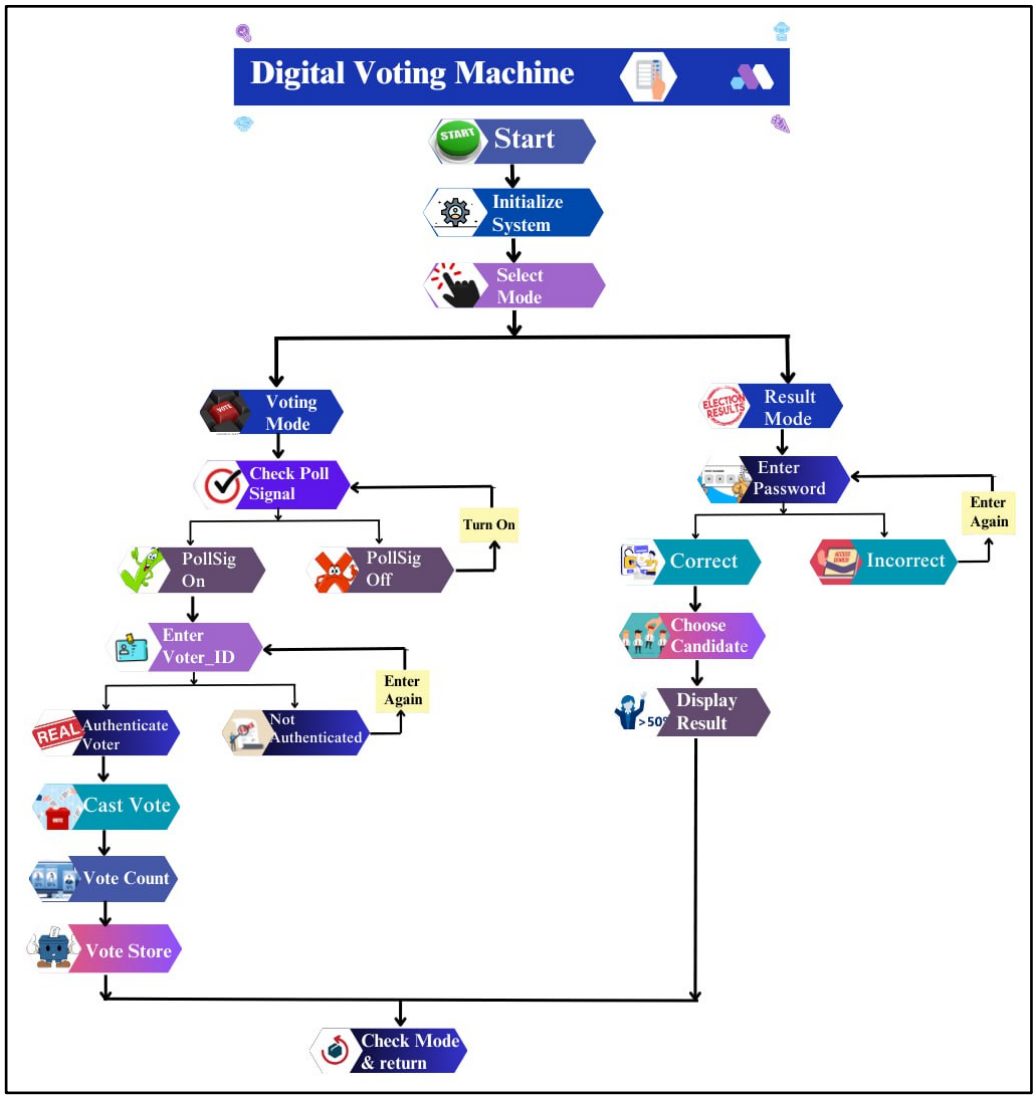


Fig: Flowchart of system design

5 Design Analysis and Evaluation

5.1 Novelty

The Digital Voting Machine (DVM) using FPGA introduces several innovative features that set it apart from traditional voting systems. By leveraging the reconfigurable and parallel processing capabilities of FPGAs, the system achieves real-time vote processing, enhanced speed, and hardware-level security. Unlike microcontroller-based systems, the FPGA-based design ensures tamper-proof vote encryption, fault tolerance, and built-in redundancy for error detection and correction. Its scalable and customizable architecture allows easy integration of advanced features like biometric authentication, multi-language support, and adaptability for various voting protocols. The machine is also energy-efficient, making it suitable for resource-constrained environments or remote locations. Furthermore, the design emphasizes transparency and reliability, with secure vote storage, auditability, and real-time tallying to reduce risks of tampering or delays. This project offers a robust, accessible, and future-ready solution to address modern electoral challenges.

5.2 Design Considerations (PO(c))

5.2.1 Considerations to public health and safety

- **Environmental Benefits**: By replacing paper ballots, the EVM helps conserve natural resources and significantly reduces paper waste, contributing to a greener electoral process.
- **Labor Efficiency**: The semi-automated nature of the EVM requires fewer personnel for election operations, minimizing manual intervention and reducing the risk of human error.
- **Increased Security and Safety**: EVMs provide advanced security mechanisms that decrease the risk of vote manipulation and tampering, promoting a safer and more reliable election environment. This also helps prevent conflicts or violence often associated with disputed results.
- **Accessible Voting Experience**: Designed for ease of use, EVMs minimize voter wait times and improve accessibility, particularly for seniors and those with disabilities, ensuring an inclusive and efficient voting process.
- **Contactless Voting Options**: The EVM can incorporate touchless or sanitized interfaces, reducing the spread of germs in crowded polling stations, an important consideration during pandemics or outbreaks.

5.2.2 Considerations to environment

- **Low Energy Consumption**: Standard EVMs are powered by batteries with minimal energy requirements, making them an eco-friendly alternative to energy-intensive traditional voting methods.
- **Reduction in Paper Waste**: By eliminating the need for paper ballots, EVMs

significantly reduce the amount of paper waste generated during elections, contributing to environmental sustainability.

- **Waste Management Benefits**: Traditional elections produce substantial waste from discarded paper ballots and ballot boxes, which require significant energy and resources for disposal. EVMs minimize this waste, reducing the environmental impact.
- **Preservation of Forest Resources**: The production of paper for ballots necessitates cutting down large numbers of trees. Transitioning to EVMs helps conserve forests and supports biodiversity.
- **Reduction of Toxic Emissions**: The inks used in paper ballots often contain heavy metals and other toxic pigments that can leach into the environment, causing soil and water contamination. EVMs avoid these hazards entirely.
- **Lower Carbon Footprint**: The lifecycle of EVMs, from production to usage, generates a smaller carbon footprint compared to the ongoing production and disposal of paper ballots.
- **Sustainable Manufacturing**: Modern EVMs can be designed using recyclable and environmentally friendly materials, ensuring that their end-of-life impact is minimal.
- **Reusable Technology**: Unlike paper ballots, which are single-use, EVMs are durable and reusable for multiple election cycles, reducing the demand for new resources.
- **Reduced Transportation Impact**: EVMs are compact and require less logistical effort compared to transporting large volumes of paper ballots and ballot boxes, cutting down on fuel usage and emissions during elections.
- **Support for Green Policies**: By adopting EVMs, electoral systems align with global efforts to promote green technologies and reduce reliance on resource-intensive practices.

5.2.3 Considerations to cultural and societal needs

- **Encouraging Youth Participation**: Younger voters are often discouraged by lengthy wait times and cumbersome identity verification processes, leading to decreased voter turnout. The proposed EVM addresses these concerns by offering a streamlined and efficient voting experience, which could reinvigorate interest and participation among the youth.
- **Restoring Public Confidence**: In regions like Bangladesh, where local elections often face allegations of vote manipulation, a secure and transparent EVM can rebuild trust in the electoral system, fostering greater faith in democratic processes among the populace.
- **Simplifying the Voting Experience**: By offering a user-friendly interface and reducing procedural complexities, the EVM makes voting more accessible and appealing to people of all age groups and education levels.

5.3 Investigations (PO(d))

The investigation of the FPGA-based Digital Voting Machine (DVM) focused on evaluating its technical feasibility, security, social impact, and environmental benefits. From a technical perspective, the project explored the integration of FPGA hardware with secure vote encryption, real-time tallying, and a user-friendly interface to improve the voting process. Key considerations included ensuring the system's reliability and scalability for various election sizes, providing robust protection against tampering, and minimizing human error. Extensive testing was conducted on the system's functionality, including stress tests to simulate real-world conditions and usability tests to ensure accessibility for all voters, including the elderly and disabled.

Socially and environmentally, the project aimed to address voter apathy by offering a streamlined and efficient system, particularly for younger voters, and by rebuilding public trust in electoral processes, especially in areas with a history of vote manipulation. Environmentally, the DVM reduces paper waste and energy consumption compared to traditional methods, promoting sustainability. The investigation also highlighted the cost-effectiveness and scalability of the system, with potential for widespread implementation across various regions and election types. The findings confirmed that the FPGA-based system is a viable, secure, and eco-friendly alternative to current voting technologies.

5.4 Limitations of Tools (PO(e))

We had to execute the whole model in an FPGA device, that's why there are some limitations. There are security and privacy problems, which will not occur in real life. When this model is used in larger scale, the poling switch, the voting or result switch, the display and the ID switch are to be to the poling officer. That's how the limitations are to be solved

5.5 Impact Assessment (PO(f))

5.5.1 Assessment of Societal and Cultural Issues

Elections are fundamental to democracy, requiring technologies that ensure transparency, accuracy, and accessibility. Electronic Voting Machines (EVMs) have improved electoral processes, and FPGA-based EVMs offer significant advancements in security, speed, and customization. However, societal and cultural factors play a crucial role in their adoption and success, including trust, inclusivity, education, accessibility, and the balance between technological progress and public acceptance.

- **The Role of FPGA in Modern EVMs**: FPGAs provide flexibility, enabling enhanced encryption, real-time data validation, and addressing security concerns. Despite these advantages, societal factors impact the system's success.

- **Trust and Transparency**: Trust in electronic voting is vital. FPGA-based EVMs can enhance transparency through secure encryption and audit trails, but the complexity of these systems may still create skepticism, particularly in societies with fragile trust in government.

- **Inclusivity and Cultural Sensitivities**: FPGA-based EVMs can support multiple languages and user interfaces, improving accessibility. However, in cultures with deeply ingrained traditional voting methods, resistance to adopting new technology may arise, particularly among elderly or rural voters.

- **Accessibility**: FPGA-based EVMs can include features for individuals with disabilities, such as braille interfaces and audio guides. However, the cost of these features may be a barrier in economically disadvantaged areas.

- **Security and Ethical Concerns**: While FPGA-based systems offer strong security, they are not immune to tampering. Regular audits, third-party verification, and stringent security protocols are necessary to maintain public trust and prevent misuse.

- **Bridging the Urban-Rural Divide**: Rural areas, often lacking technological infrastructure, may face challenges in adapting to FPGA-based EVMs. Mobile units and on-site education can help address these disparities and encourage broader participation.

- **Political and Institutional Challenges**: Resistance from political actors, especially those fearing bias, can impede the adoption of FPGA-based EVMs. Transparent procurement and bipartisan oversight are essential to gaining public confidence.

5.5.2 Assessment of Health and Safety Issues

- **Physical Health Concerns**: Long hours of use can lead to discomfort or repetitive strain injuries (RSIs) for both voters and poll workers. Ensuring ergonomic design, such as adjustable screens and tactile feedback, can reduce physical strain.
- **Stress And Anxiety:** Both poll workers and voters may experience stress related to operating the machines. Comprehensive training for election staff and public awareness campaigns can mitigate this psychological pressure.
- **Accessibility and Health Equity**: FPGA-based EVMs must accommodate voters with disabilities, providing features like braille and audio guides. Ensuring uniform accessibility across polling stations is key for equity.

5.5.3 Assessment of Legal Issues

- **Voter Data Security**: FPGA-based EVMs process sensitive voter data, requiring compliance with data protection laws like GDPR. Strong encryption and secure storage are necessary to protect voter information.

- **Cross-Border Data Transfers**: In multinational elections, voter data may cross borders, subject to complex international laws. Strong encryption and legal agreements are necessary to comply with these regulations.

- **Electoral Commission Responsibility**: Electoral commissions are legally responsible for ensuring EVM compliance with all laws, including certification, transparency, and accessibility.

- **Legal Mandates for Transparency**: Legal frameworks require EVMs to be transparent, allowing independent audits and public scrutiny while balancing security concerns.

5.6 Ethical Issues (PO(h))

The adoption of FPGA-based Electronic Voting Machines (EVMs) raises significant ethical concerns that must be addressed to ensure their alignment with democratic principles and public trust. Key issues include the lack of transparency due to the complexity of FPGA technology, which can lead to public skepticism and mistrust. Balancing proprietary systems with open-source solutions is essential for enabling independent audits while maintaining security. Additionally, ensuring accessibility for marginalized groups, protecting voter autonomy, and safeguarding privacy and anonymity are critical ethical considerations. The risk of manipulation, voter coercion, and tampering, as well as data breaches, must be mitigated through strong security measures. Ethical deployment also requires addressing the digital divide, language, and cultural sensitivity, ensuring equitable access, and preventing over-reliance on automation, which could diminish human oversight. Developers, governments, and civil society must collaborate to ensure that FPGA-based EVMs are deployed fairly, securely, and transparently, with mechanisms in place for accountability and dispute resolution. Long-term impacts, including the erosion of traditional electoral knowledge, must also be considered. Ultimately, ethical innovation guided by principles of transparency, fairness, and inclusivity is necessary to maintain public trust and uphold the integrity of the electoral process.

6 Reflection on Individual and Team work (PO(i))

6.1Individual Contribution of Each Member

Each team member brought unique feature to this project.

Student_ID	Contribution
2006086, 2006098	Base Module, Mode Control & Button Control (Vote casting and Result mode) Code Development
2006083, 2006084	Valid Vote Counting & Voter ID Authentication Code Development
2006083, 2006098, 2006086	Vote Display, Password System Code & Combining all Process.
2006084, 2006086, 2006098	Hardware Implementation on UP2 Board, Pin Planning & Operating FPGA.
2006083, 2006084, 2006086, 2006098	Presentation Slide and Report Writing

6.2 Mode of Team Work

The development of this electronic voting machine (EVM) system required a collaborative and multidisciplinary approach, leveraging the strengths of team members with diverse expertise. The project was executed in clearly defined phases, including system design, hardware integration, software development, and testing. Each phase was assigned to sub-teams based on their specialization. For instance, hardware engineers focused on integrating components like the UP2 board, Bluetooth module, and 7-segment displays, while software developers worked on implementing the Verilog code for functionalities such as vote logging, voter authentication, and result display. Regular meetings and progress reviews ensured synchronization between sub-teams, fostering seamless integration of hardware and software components. Communication and task management tools were utilized to track deliverables and address challenges collaboratively. Additionally, cross-functional brainstorming sessions were held to refine features and explore future scalability options, highlighting the importance of teamwork in delivering a robust and efficient system.

6.3 Diversity Statement of Team

Our team values diversity in skills, perspectives, and experiences, certain team members demonstrated a higher level of proficiency in Verilog coding while others possessed a deeper knowledge of the subject matter to set algorithm which was essential for the success of this project. Consequently, we allocated our responsibilities in accordance with our respective strengths. By combining expertise in hardware, software, and system design, we were able to address challenges creatively and deliver a well-rounded solution. This inclusive approach fostered innovation, ensured comprehensive problem-solving, and enhanced the overall quality of the electronic voting machine system. Ultimately, the

successful realization of the project was made possible through the accumulation of our diverse skill sets, leading to the attainment of PO(i).

6.4 Log Book of Project Implementation

Our timeline includes the following crucial phases:

Date	Milestone Achieved
22 September -23 September	Team meeting and listing project for proposal.
24 September	Final proposal submission
27 September - 13 October	Going through research paper and previous works related to our project.
25 October - 15 November	Verilog code development.
16 November - 20 November	Hardware implementation and Pin planning.
23 November - 26 November	Debugging and Code adjustment.
27 November - 15 December	Report writing.

7 Communication to External Stakeholders (PO(j))

7.1 Executive Summary

***Press Release**: Revolutionizing Elections with FPGA-based Digital Voting Machines*

A groundbreaking Digital Voting Machine has been developed using advanced Field Programmable Gate Array (FPGA) technology. Designed on the Altera UP2 Development Board, this system ensures secure, error-free voting and real-time result display. By leveraging mathematical computations, voter authentication, and efficient vote counting mechanisms, it eliminates risks of multiple voting and human errors.
Key features include secure voter ID verification, fast vote logging, and a 7-segment display for instant results. With its high speed, flexibility, and reliability, this innovative system redefines modern election processes, promising a secure and transparent future for digital voting.

7.2 User Manual

User Manual for Digital Voting Machine

Introduction

This manual provides a step-by-step guide to operate the Digital Voting Machine (DVM) implemented on an FPGA. The system is designed for secure and accurate electronic voting with real-time vote counting and display.

System Components

The following hardware components are included in the system:
FPGA Board: The core of the voting machine where the logic is implemented.
Input Buttons/Switches: Used to cast votes for candidates.
7-Segment Display/LEDs: Displays the vote counts for each candidate.
Power Supply: Provides power to the FPGA and peripherals.
Reset Button: Resets all counters and initializes the voting session.

Getting Started

Powering On

- Connect the FPGA board to the power supply.
- Ensure all connections to input devices (buttons) and output displays (7-segment/LCD) are secure.
- Turn on the power. The system will initialize and display "0" on all counters.

Casting Votes

Voting Process

1. Each candidate is assigned to a specific button or switch.
2. Press the corresponding button for 3 seconds to cast a vote for the desired candidate. Keep pressing till the `already_voted` LED turns on.
3. After pressing the button:
    - The system will register the vote.
    - The counter for the selected candidate will increment by one.
    - The updated vote count will be displayed on the output device.

Single Vote Per User

- The system is designed to allow one vote per voter. Ensure proper monitoring during voting to prevent unauthorized multiple voting.

Ending the Voting Session

Displaying Results

- Once all votes are cast, the system displays the total votes for each candidate on the output device.
- Use the "result button" or pre-programmed trigger to finalize the tally.

Resetting the System

- To start a new voting session, press the reset button.

- This will reset all vote counts to zero, and the system will be ready for the next voting cycle.

Troubleshooting

FAQ and Solutions

- **Issue**: No response when a button is pressed.
  Solution: Check the button connection to the FPGA. Ensure no loose wires.
- **Issue**: Vote counts not displayed correctly.
  Solution: Verify the display connections and ensure proper initialization at startup.
- **Issue**: System doesn't reset properly.
  Solution: Press the reset button firmly. If the problem persists, restart the FPGA board.

Safety Precautions

- Ensure the system is operated in a controlled environment.
- Avoid exposing the FPGA board to moisture or extreme temperatures.
- Handle the components carefully to prevent physical damage.

7.3 Drive Link of Project Code

https://drive.google.com/drive/folders/1sZIhoGiN8SgVodzSXr69AEJOJlsSZpm3?usp=sharing

7.4 YouTube Link

https://youtu.be/mCLIivEqHV0?si=t_kxi9vghe5bbm9h

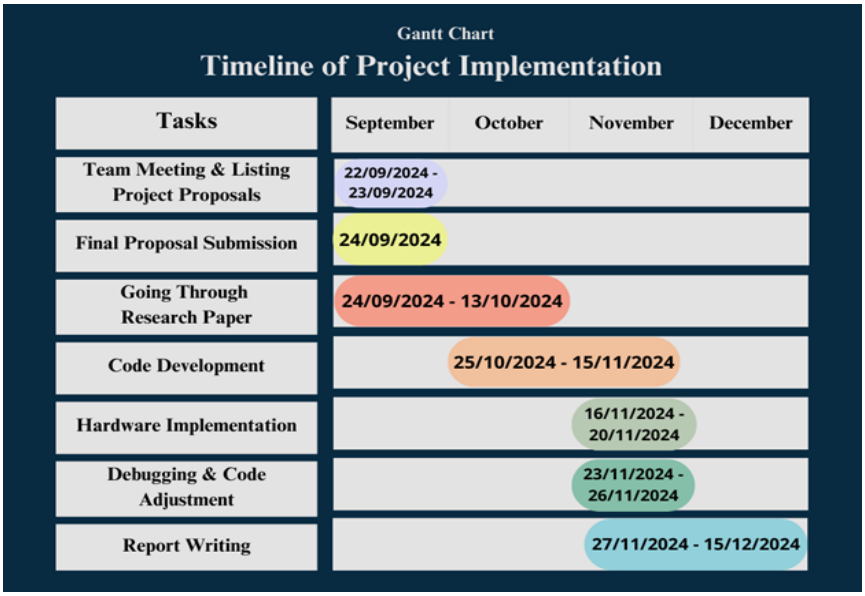
8 Project Management and Cost Analysis (PO(k))

8.1 Bill of Materials

Cost Item	Estimated Cost
Hardware Procurement ALTERA FPGA Board HC-05 Module jumper wire set Breadboard	N/A(from lab) 315/= 100/= 100/=

Software Development Tools Quartus licensing	N/A (from lab)
Total Estimated Cost	515/=l

8.2 Timeline of Project Implementation



9 Future Work (PO(l))

Scalability for Larger Applications: The current system, designed for small-scale use, can be expanded to accommodate a larger number of voters and candidates. This will involve optimizing the design and increasing the capacity of the voter ID storage and processing units.

Improved Voter ID Input System: To make the voter ID input process more user-friendly and efficient, a **Bluetooth module** (e.g., HC06 or HC05) will be integrated into the system. This will enable voters to input their IDs using a mobile phone application, reducing manual input errors and enhancing the overall voting experience.

Data Backup and Recovery: The system can include a **backup mechanism** to save voting data periodically to an external storage device or cloud platform. In case of a system failure,

data recovery will ensure continuity without losing previously cast votes.

Dynamic Candidate Management: The current implementation hardcodes the number of candidates and their respective vote counts. Future updates can include a **dynamic configuration system**, allowing election administrators to define the number of candidates and their identifiers during the setup phase.

Enhanced Digit Display: The current two-digit display for vote counts will be upgraded to **more digit display** to accommodate larger vote totals. This is particularly important for large-scale elections where the number of votes can exceed the two-digit range.

Improved Voter Authentication: The existing voter authentication relies on preventing double voting using a voter ID. Future system can integrate **biometric authentication** (fingerprint or facial recognition) to ensure a higher level of voter verification.

Accessibility Features for Persons with Disabilities: To make the system more inclusive, future versions can include audio guidance, tactile buttons, or voice-activated voting options to assist voters with disabilities.

10 References

https://github.com/AnitLearner/

https://www.scribd.com/document/689692983/

International Journal of Engineering & Advanced Technology (IJEAT)

A.K. Wang, R. Ligmanowski, J. Castro and A. Mazzara, "EVM Simulation and Analysis Techniques," MILCOM 2006 - 2006 IEEE Military Communications conference, Washington, DC, USA, 2006, pp. 1-7, DOI: 10.1109/MILCOM.2006.302043. keywords: {Analytical models; Transmitters; Nonlinear distortion; GSM; Wireless LAN; Military communication; Physical layer; Phase distortion; Phase modulation; Phase noise}